Vaishnavi Kashyap

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Full-stack engineer with production manufacturing-systems experience, now building agentic AI systems and RAG pipelines with the Anthropic SDK. M.S. Computer Science, GPA 4.0, UMass Amherst, May 2026. [portfolio]

EXPERIENCE

Office & Technology Operations Intern **Aug 2026 – Present**
Hometown Nannies (Boomerroomers) *Remote*

- Replaced inconsistent crash behavior across **8 data screens** of a React Native app with a standardized, retry-enabled error pattern built on a reusable error-boundary component; fixed the notifications screen's broken error state and flagged two more production defects via peer-reviewed pull requests.

Software Engineer **Nov 2023 – Aug 2024**
Saint Gobain *Mumbai, India*

- Replaced paper sketches and verbal coordination for 100+ factory workers by building a React + Node.js + Three.js assembly floor planner with a drag-and-drop 2D canvas, a live 3D barcode-scan status view, and AutoCAD-derived geometry, delivering **90% operational efficiency improvement**.
- Proposed and led a carbon emissions platform that replaced fragmented spreadsheet reporting across 5+ factory sites, built on **React + Django + PostgreSQL + AWS**: normalized schema design, DRF REST APIs with input validation, S3 signed-URL exports, and EC2/Gunicorn/CloudWatch deployment serving **300+ users at 99.9% uptime**.
- Cut roughly 1-week paper audit cycles to near real-time by delivering a factory audit workflow automation system (Angular + Django) with a five-state workflow backend and auto-routing of flagged non-compliance records by site and category.

Data Science Intern **Jun 2022 – Aug 2022**
CloudSufi *San Francisco*

- Owned an end-to-end ARIMA forecasting pipeline on **GCP Vertex AI** across **1M+ records** of Scope 1 emissions with weather covariates, using ACF/PACF parameter selection and holdout validation, and deployed it to production with BigQuery-connected AppSheet dashboards.
- Cut analyst query time by **70%** by designing BigQuery SQL views with multi-table joins and rolling 2-week aggregation windows; presented findings directly to the **CEO and full ML team**.

PROJECTS

Agentic F1 Race-Analytics Assistant | FastAPI, Anthropic SDK, React, PostgreSQL, Celery [github] **Jun 2026 – Aug 2026**

- Reduced silent hallucination risk in LLM sports Q&A by building an **8-tool agentic pipeline (FastAPI + Claude via the Anthropic SDK)** that grounds every answer in live OpenF1 telemetry and caps the model's self-reported confidence at **0.2 when no tool returns data and 0.6 when data is thin**, with sources rebuilt from the tools that actually ran; runs on a **5-service Docker Compose stack**.
- Shipped a CI-gated eval harness scoring **10 cases across 9 categories** for correctness, calibration, and fallback behavior, including a question about a race that doesn't exist; backed by **66 automated tests at 83.78% coverage** in a 5-job **GitHub Actions** pipeline that fails the build on any gate failure.

Legal Contract Q&A RAG System | PyTorch, RAGAS, MLflow, ChromaDB, FastAPI [github] **Apr 2026 – Jun 2026**

- Built a RAG pipeline over **510 legal contracts** (CUAD benchmark, 13K+ annotated clauses) fusing BM25 keyword search with dense retrieval via Reciprocal Rank Fusion, served via FastAPI + Streamlit in Docker; generated answers through a swappable OpenAI SDK client (Llama 3.1 while iterating, GPT-3.5-turbo for final numbers).
- Fine-tuned a PyTorch embedding model on 11K pairs to lift held-out top-1 retrieval accuracy from **3.5% to 8.0%**, evaluated with RAGAS across **30+ MLflow runs**; diagnosed a train/production granularity mismatch to land the best composite RAGAS score of 4 configs tested.

Near Real-Time Air Quality Monitoring Platform | FastAPI, PostgreSQL, Celery, Redis [github] **Feb 2026 – Jun 2026**

- Resolved a **74% v1 error rate** in a live air-quality ingestion pipeline by moving synchronous weather API calls into a Celery-populated table behind an idempotent PostgreSQL time-series store; **load-tested to 230 RPS at 100 concurrent clients with 0 errors**; 27 pytest tests and GitHub Actions CI.

EDUCATION

University of Massachusetts Amherst **May 2026**
M.S. in Computer Science | GPA: 4.0 / 4.0 | Coursework: Neural Networks, Network Security, Software Engineering, Cryptography

Vellore Institute of Technology, Vellore **May 2023**
B.Tech. in Computer Science | GPA: 3.86 / 4.0 | Coursework: Data Structures & Algorithms, Operating Systems, Database Systems

SKILLS

AI / Agentic: Anthropic SDK (tool use), Claude Code, RAG (BM25 + dense, RRF), RAGAS, MLflow, ChromaDB, PyTorch, LangChain, LLM evals
Languages: Python, JavaScript, TypeScript, SQL, C++, Bash
Web: React, React Native, Angular, Three.js, Chart.js, FastAPI, Django, DRF, Node.js, SQLAlchemy, Celery, Redis, PostgreSQL
Cloud & DevOps: AWS (S3, EC2, CloudWatch), GCP (Vertex AI, BigQuery), Docker, GitHub Actions, pytest, Agile/Scrum